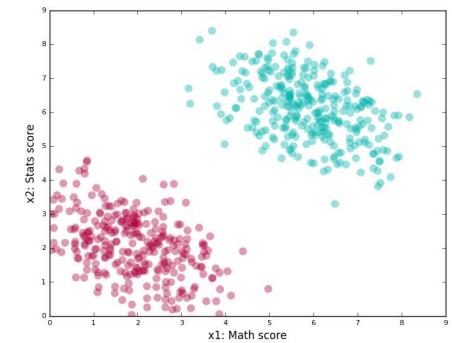


- 1 Problem of Classification
- 2 SVM
- 3 Kernel Trick
- 4 Multiclass classification

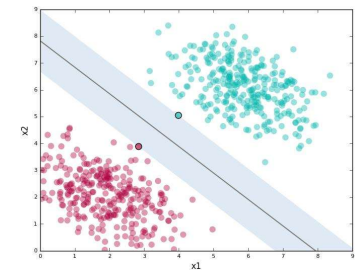
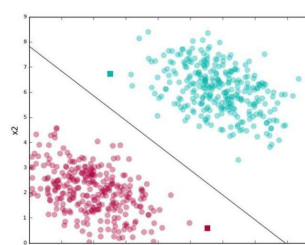
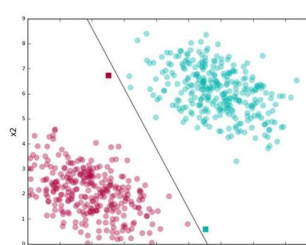
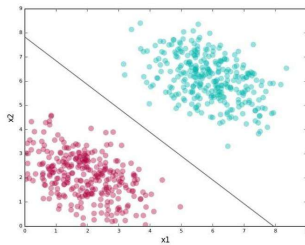
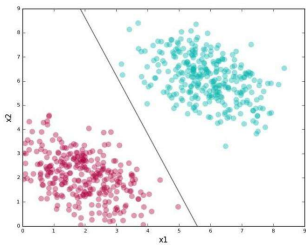


## The Problem of Classification

## The Problem of Classification

## SVM

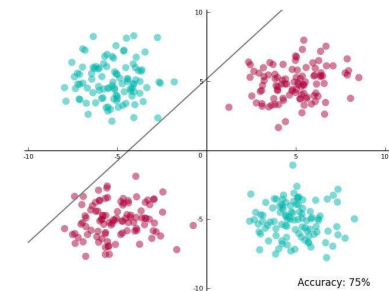
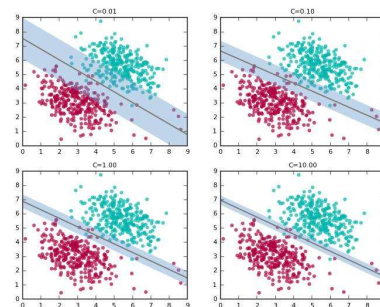
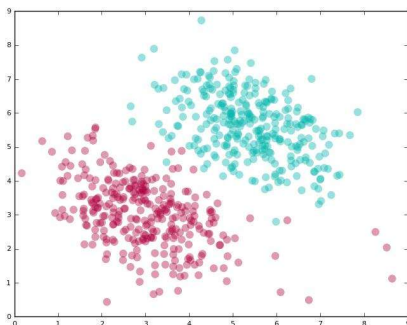
- Find lines that correctly classify the training data
- Among all such lines, pick the one that has the greatest distance to the points closest to it.

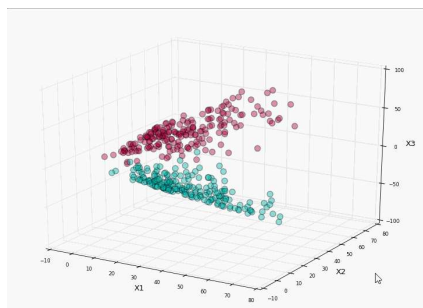


## How about errors?

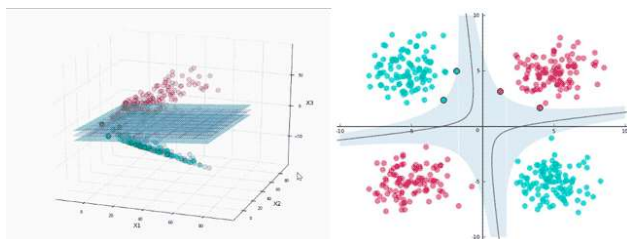
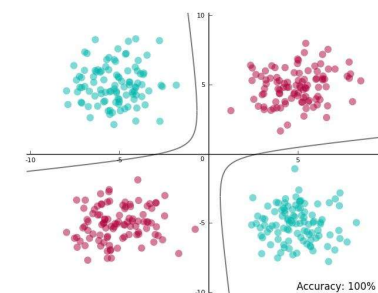
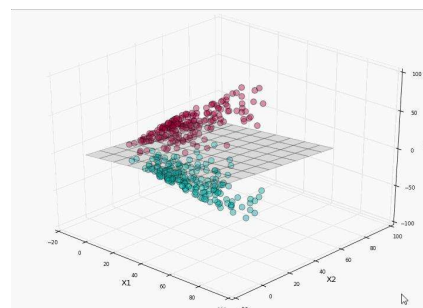
## C parameter in SVM

## Non-linearly Separable Data





$$\begin{aligned} X_1 &= x_1^2 \\ X_2 &= x_2^2 \\ X_3 &= \sqrt{2}x_1x_2 \end{aligned}$$



- One vs. One
- One vs. Rest

For example consider four class problem having classes A, B, C, and D.

- One vs. Rest
  - Models classifiers\_A, classifier\_B, classifier\_C and classifier\_D
  - During prediction here is the probability we get:
    - classifier\_A = 40%
    - classifier\_B = 30%
    - classifier\_C = 60%
    - classifier\_D = 50%
  - We assign it class C

- One vs One
  - We train total six classifier with subset of data containing classes involved
    - classifier\_AB
    - classifier\_AC
    - classifier\_AD
    - classifier\_BC
    - classifier\_BD
    - classifier\_CD
  - And during classification
    - classifier\_AB assigns class A
    - classifier\_AC assigns class A
    - classifier\_AD assigns class A
    - classifier\_BC assigns class B
    - classifier\_BD assigns class D
    - classifier\_CD assigns class C
  - We assign it to class A

- SVM for classification
- Kernel tricks
- Multiclass classification

Cảm ơn đã theo dõi!